Task-21:- Requirement Analysis

1. Introduction

1.1 Purpose

The purpose of this document is to define the requirements for deploying a **Hello World application on Azure App Service**, configuring a **custom domain**, and validating DNS records including **CNAME, TXT, and SOA records**.

This document is intended for:

- DevOps Engineers / Interns
- Cloud Engineers
- Project Stakeholders
- Evaluators / Reviewers

1.2 Scope

In Scope

- Creation of Azure App Service
- Deployment of Hello World application
- Configuration of custom domain
- DNS record setup (CNAME and TXT)
- DNS validation using command-line tools
- SOA record inspection and explanation

Out of Scope

- SSL certificate configuration (HTTPS)
- CI/CD pipeline setup
- Complex application deployment
- Load balancing and scaling

1.3 Definitions, Acronyms, and Abbreviations

Term	Description
Azure App Service	Platform to host web applications

DNS	Domain Name System
CNAME	Canonical Name record used for domain mapping
TXT Record	Text record used for verification
SOA	Start of Authority record
FQDN	Fully Qualified Domain Name
nslookup	Tool to query DNS records
DevOps	Development and Operations

1.4 References

- Microsoft Azure Documentation
- DNS Fundamentals Documentation
- Domain Provider (GoDaddy/Namecheap) Guides
- RFC 1035 (DNS Standard)

1.5 Overview

This document provides a structured understanding of the project including:

- Project objectives and business goals
- Functional and non-functional requirements
- Use cases and system behavior
- Risks, assumptions, and dependencies

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2. Project Overview

2.1 Project Objectives

- Deploy a Hello World application on Azure
- Configure and validate a custom domain
- Implement DNS records (CNAME and TXT)
- Analyze SOA record fields
- Ensure application accessibility via custom domain

2.2 Business Goals

- Enable cloud-based application hosting
- Improve understanding of DNS infrastructure
- Ensure domain ownership verification
- Provide scalable and accessible application hosting

2.3 Background

ABC Technologies aims to migrate its applications to cloud infrastructure using Microsoft Azure. This project helps the team understand:

- Azure App Service deployment
- Domain mapping
- DNS record management
- DNS zone configuration

2.4 Constraints

- Limited time (3 hours completion target)
- DNS propagation delay
- Dependency on domain provider
- Azure service availability

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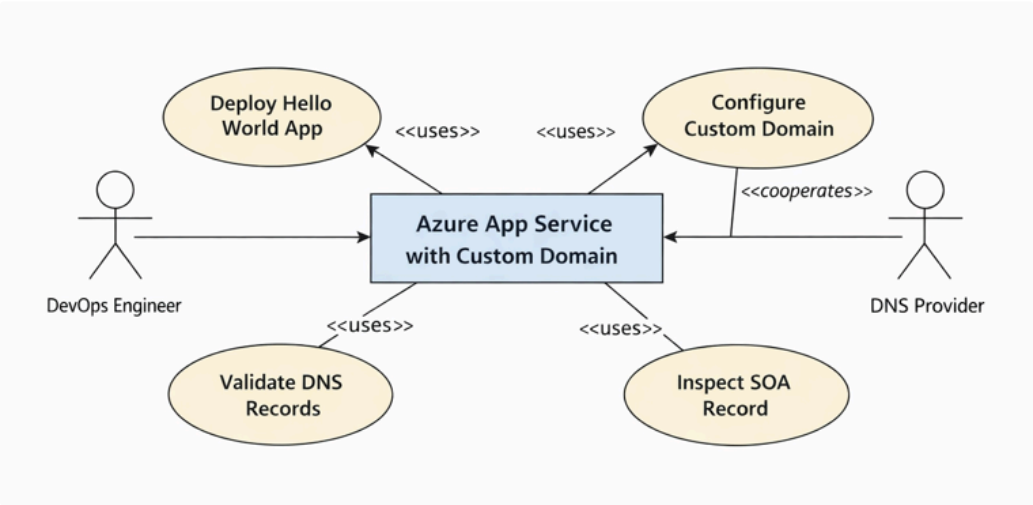
3. Functional Requirements

3.1 Use Case Diagram



3.2 Use Cases

3.2.1 Use Case 1: Deploy Hello World Application

Description:

Deploy a basic Hello World application on Azure App Service.

Actors:

- DevOps Engineer

Preconditions:

- Azure account is available
- User is logged into Azure portal

Postconditions:

- Application is deployed and accessible via default Azure URL

Flow of Events:

1. User logs into Azure Portal
2. User creates App Service
3. User deploys Hello World application
4. Azure hosts the application
5. User accesses application via default URL

Alternative Flows:

- Deployment fails → Retry deployment
- App not loading → Restart service

3.2.2 Use Case 2: Configure Custom Domain

Description:
Map a custom domain to the Azure App Service.

Actors:

- DevOps Engineer
- DNS Provider

Preconditions:

- App Service is running
- Domain name is available

Postconditions:

- Custom domain is successfully mapped

Flow of Events:

1. User navigates to Custom Domain section in Azure
2. User adds domain name
3. User creates CNAME record in DNS provider
4. User creates TXT record for verification
5. Azure validates ownership
6. Domain is linked successfully

Alternative Flows:

- DNS validation fails → Check TXT record
- Domain not resolving → Wait for propagation

3.2.3 Use Case 3: Validate DNS Records

Description:
Verify DNS records using command-line tools.

Actors:

- DevOps Engineer

Preconditions:

- DNS records are created

Postconditions:

- DNS resolution is verified

Flow of Events:

1. User runs `nslookup` for CNAME
2. User checks TXT record
3. User verifies domain resolution
4. User confirms app accessibility

Alternative Flows:

- DNS not resolving → Wait and retry

3.2.4 Use Case 4: Inspect SOA Record

Description:
Analyze the SOA record of the DNS zone.

Actors:

- DevOps Engineer

Preconditions:

- Domain is active

Postconditions:

- SOA record fields are documented

Flow of Events:

1. User runs DNS query for SOA record
2. System returns SOA details
3. User documents each field
4. User explains each parameter

Alternative Flows:

- No response → Check domain or DNS

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4. Non-Functional Requirements

4.1 Performance Requirements

- Application should load within 2–3 seconds
- DNS resolution should be quick (<1 second typical)

4.2 Security Requirements

- Domain ownership verification via TXT record
- Secure access to Azure portal
- No unauthorized domain mapping

4.3 Usability Requirements

- Simple and user-friendly Azure interface
- Clear DNS configuration steps
- Easy verification process

4.4 Reliability Requirements

- High availability of Azure App Service
- DNS should resolve consistently
- Minimal downtime

4.5 Compliance Requirements

- Must follow DNS standards (RFC 1035)

- Must comply with Azure service policies

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5. Assumptions and Dependencies

5.1 Assumptions

- User has Azure subscription
- Domain name is owned by user
- DNS provider supports record updates
- Stable internet connection

5.2 Dependencies

- Azure App Service availability
- Domain provider services
- DNS propagation
- Network connectivity

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6. Risk Analysis

6.1 Risk Identification

Risk	Impact
DNS propagation delay	Domain not accessible
Incorrect DNS configuration	Mapping failure
Azure validation failure	Domain linking fails
App deployment failure	Application not available

6.2 Risk Mitigation

Risk	Mitigation
DNS delay	Wait and verify later
Wrong DNS records	Double-check entries

Validation failure	Ensure TXT record accuracy
Deployment issues	Redeploy app

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7. Glossary

Term	Meaning
Azure App Service	Cloud hosting platform
CNAME	Domain alias record
TXT Record	Verification record
SOA	DNS zone authority record
DNS	Domain resolution system

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8. Appendices

- Screenshots of Azure App Service
- Hello World Application
- DNS Records (CNAME & TXT)
- Custom Domain Working
- SOA Record Output

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9. Customer Acknowledgement

Name: _____

Title: _____

Signature: _____

Date: _____